

Simulating a hanging cable: mathematics, measurement, and three solvers compared

Generated by `analysis/make_thesis_report.py`

July 21, 2026

Abstract

A flexible cable is suspended from 3 level clamps spanning $S = 0.350$ m and photographed. Its shape is predicted three independent ways: from the governing differential equation, from a set of physics solvers, and from the photograph itself. The three-way structure is deliberate — any two of them can agree for uninteresting reasons, and only the third makes it possible to attribute a disagreement to numerical error rather than to modelling error. This document reports what agrees, what does not, and which results are currently trustworthy.

Summary of findings

- The governing equation, integrated numerically without recourse to its closed form, reproduces $z = a \cosh((x - x_0)/a) + c$ to $7 \times 10^{-3} \mu\text{m}$. The analytic reference every error here is measured against is therefore verified rather than assumed.
- Newton cable - point pin (kink) reproduces the analytic catenary to 0.91 mm on a 893 mm cable (0.26% of the span).
- Agreement with the mathematics and agreement with the photograph rank the solvers in opposite orders. The gap measures the difference between a clamped and a pinned end, not solver accuracy.
- Imposing the real (flat-tape) middle boundary condition rounds the kink and moves the Newton cable from 75 mm to 65 mm against the photograph, while raising its error against the idealised catenary from 0.9 mm to 21 mm — confirming the residual is a boundary condition, not solver error.
- Solver accuracy depends on the PRODUCT of substeps and iterations, but cost does not: each substep carries fixed overhead an iteration does not, so fewer substeps with more iterations is strictly cheaper at equal accuracy.
- Damping the stretch constraint is actively harmful — it opposes the solver’s own length correction — while damping the bending constraint does nothing measurable across a $600 \times$ range.
- The Warp rod is the only method imposing a true position-only pin, so its shape depends on the one input this study cannot measure: the bending stiffness. Its error against the catenary rises from 53 mm at the cable’s drape-consistent $\ell \approx 20$ mm (used here) to 80 mm at the stand-in material’s $\ell \approx 150$ mm. At coarse resolution the stiff case additionally buckles into an arch 150 mm above the supports; refining the mesh removes the buckle, so that was a discretisation artefact while the stiffness sensitivity is not.

1 Apparatus

The cable is clamped with tape to a vertical door at 0.000 m, 0.150 m, 0.350 m measured from the leftmost clamp, all at one height, and photographed square-on. Cable length is not taken from the packaging: the free length between clamps is what governs the shape, so it is measured from the same photograph the simulations are compared against. Here that gives $L = 0.8932$ m, and $L/S = 2.552$ — a very slack cable, sagging further than it spans.



Figure 1: The cable as photographed. The tape fixes the cable’s **ANGLE** as well as its position, which is a clamped boundary condition, not the pinned one the classical catenary assumes.

2 Image processing

The cable is separated from the door in HSV: it is bright and nearly unsaturated where the door is a saturated brown. The connected component is then selected by **SHAPE** rather than by area — a specular highlight on the door is also bright and unsaturated, and is often *larger* than the cable, so selecting the largest region finds the glare instead. A cable is thin and spans the clamps; glare is a blob.

The centreline is then traced **ALONG** the curve: from each point the tracer advances one step along the local tangent and re-centres on the mask’s perpendicular cross-section. The obvious alternative — averaging the mask rows in each pixel column — fails badly here, because a slack cable hangs near-vertically close to its clamps and one column then spans a long run of cable.

Figure 2 is the check on all of this: the left panel tints every segmented pixel so a mis-segmentation is visible directly, and the right panel compares the extracted profile to a fitted catenary, because a clean segmentation can still fit badly and those are different failures.

Cable detection: three_point.jpg

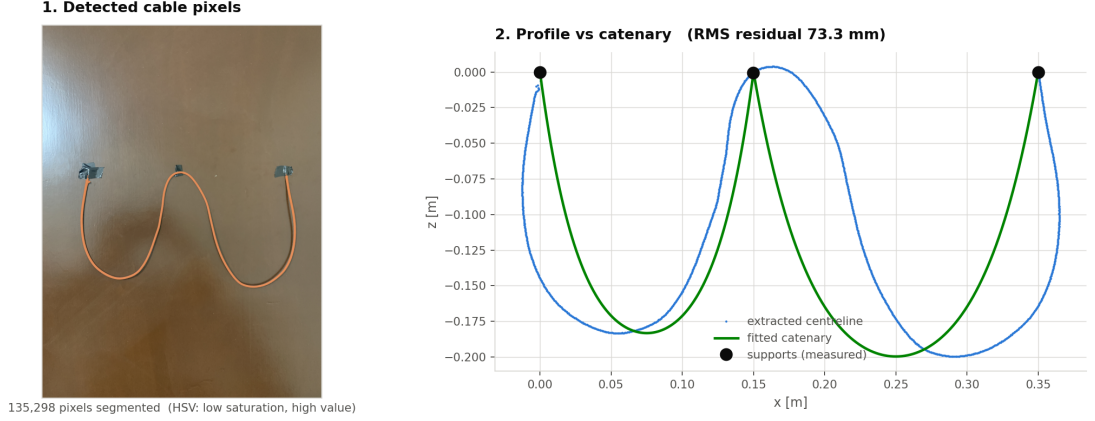


Figure 2: Detection and fit. Left: every segmented pixel, tinted over the original. Right: extracted profile against the fitted catenary.

3 Mathematical model

For a perfectly flexible, inextensible cable of weight w per unit length, the horizontal component of tension is constant along the cable, $T \cos \theta = H$. Balancing the vertical forces on an element ds and substituting $ds = \sqrt{1 + z'^2} dx$ gives the governing equation

$$z'' = \frac{\sqrt{1 + z'^2}}{a}, \quad a = \frac{H}{w}, \quad (1)$$

a two-point boundary-value problem closed by the arc-length constraint $\int_0^S \sqrt{1 + z'^2} dx = L$, which is what fixes a . Physically a is set by how much cable was hung: a slacker cable has a smaller a and therefore carries *less* tension.

The corresponding dynamics, which is what the solvers actually integrate, is

$$\rho A \frac{\partial^2 \mathbf{r}}{\partial t^2} = \frac{\partial}{\partial s} \left(T \frac{\partial \mathbf{r}}{\partial s} \right) + \rho A \mathbf{g}, \quad (2)$$

with $|\partial \mathbf{r} / \partial s| = 1$ enforced by T , which is a Lagrange multiplier rather than a material property. Setting $\partial^2 \mathbf{r} / \partial t^2 = 0$ recovers (1), which is the assumption that licenses scoring a settled simulation against the static solution.

Verification. Equation (1) is integrated numerically (RK4 from the vertex, bisection on a for the length constraint) without using the closed form $z = a \cosh((x - x_0)/a) + c$ anywhere. The two agree to $7 \times 10^{-3} \mu\text{m}$. The analytic reference every error in this document is measured against is therefore verified rather than assumed.

4 Simulation

Each solver runs in its own process — Isaac Sim's **SimulationApp** is a singleton and the methods force different physics engines — and each is handed the identical scenario file, so a method that disagrees is disagreeing about physics rather than about its inputs.

method	RMSE [mm]	max [mm]	sag [mm]	arc drift [%]	settled	wall [s]
Newton cable - point pin (kink)	0.91	2.94	220.7	0.855	yes	12.9
Newton cable - tape (rounded)	21.12	72.65	218.3	1.064	yes	11.7
PhysX capsule chain (D6)	74.40	150.39	204.0	0.457	NO	29.0
Warp XPBD rod	44.78	96.60	212.4	0.133	yes	1.3

Table 1: Each solver against the analytic catenary. Arc drift is the reference-free check: an inextensible cable must keep its length whatever the catenary says.

Boundary conditions are not a detail. These runs do not all hold the middle support the same way, and the choice moves the shape more than the solver does. The Newton cable appears twice, as the same solver under two boundary conditions: *point pin* welds the cable at the catenary’s own angle, which corners into a kink; *tape* holds a short flat plateau, so the cable leaves horizontally and its bending rounds the top, as a real strip of tape does. The PhysX and Warp methods pin the node and let the cable pivot. The point pin is nearest the analytic curve and the tape nearest the photograph — the same trade-off the conclusions turn on (Section 8) — so the table compares *models*, not solver quality. The middle clamp was measured to hold 46.9% of the cable’s arc length on its left, where the chord-proportional convention assumes 42.9% (+4.0 percentage points). Since the clamp pins a material point, this split is a property of the apparatus rather than something statics determines, so it must be measured and imposed.

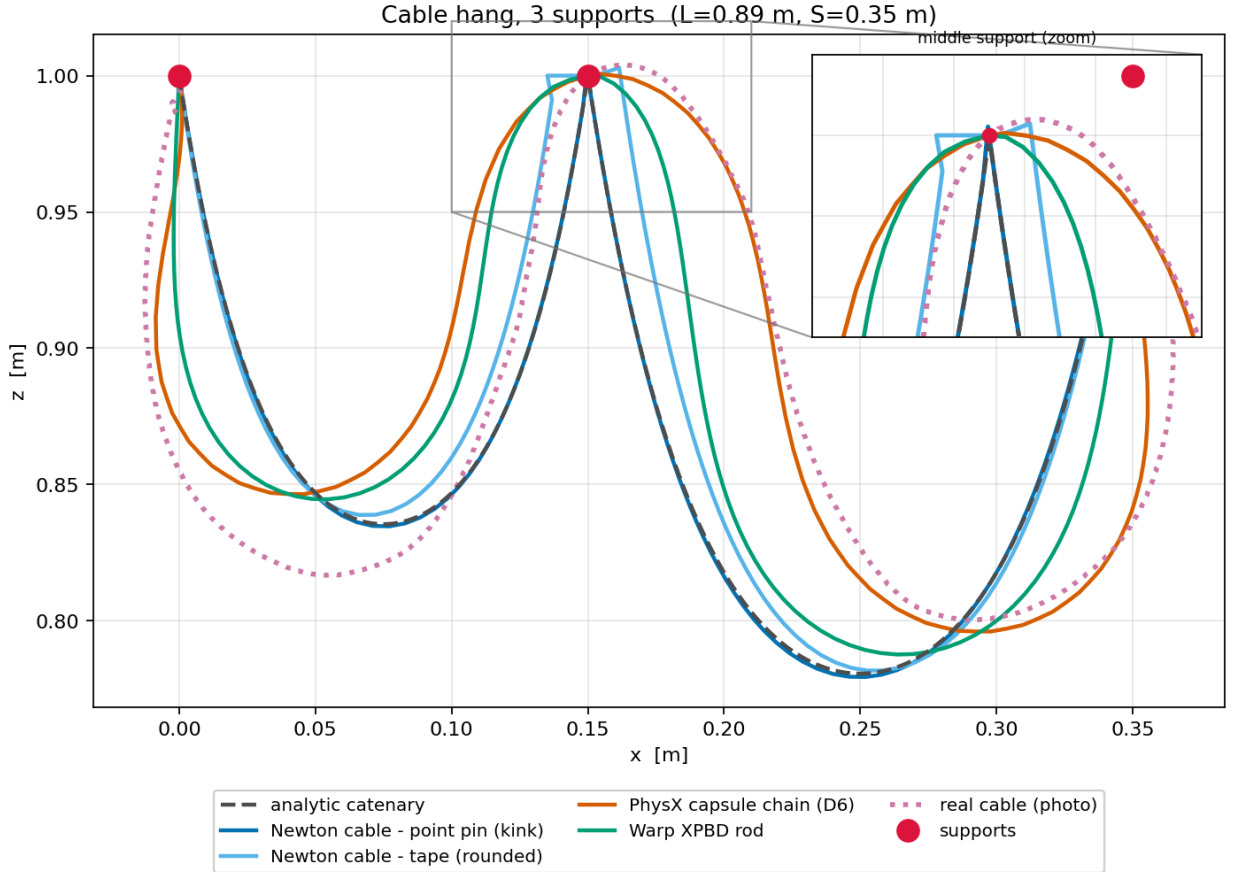


Figure 3: All methods, the analytic catenary, and the photographed cable on one axis. Inset: the middle support magnified. The point pin and the analytic catenary corner in an upward kink; the flat-tape model and the real cable both round over. That contrast is the boundary-condition result of Section 8 made visible.

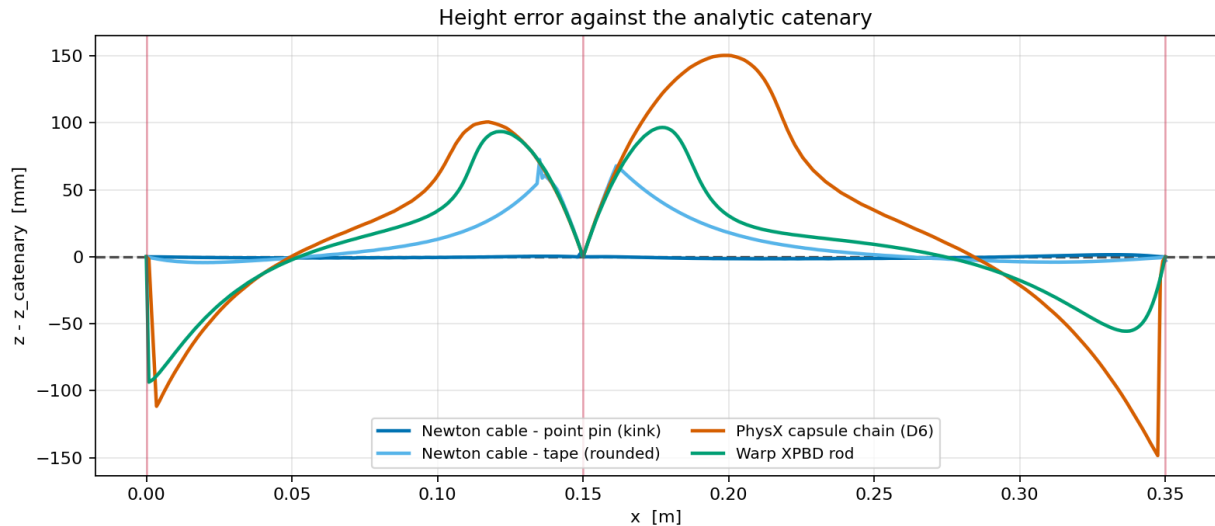


Figure 4: Deviation from the analytic catenary along the cable.

method	RMSE vs photo [mm]	max [mm]
Newton cable - point pin (kink)	75.10	159.59
Newton cable - tape (rounded)	65.31	159.20
PhysX capsule chain (D6)	32.77	156.36
Warp XPBD rod	51.16	151.45

Table 2: Simulation against the photographed cable. The extraction’s frame is calibrated on the two outer clamps, which are level in the apparatus; the middle clamp is recovered to within 14 mm as an independent check (Section 7), which bounds the accuracy of these figures.

5 Solver hyperparameters

Robustness across slackness. The sweep was run at two geometries — $L/S = 1.25$ and the photographed $L/S = 2.55$, twice as slack — and the conclusions are the same in both. Newton’s default reaches 27.8 mm at the lower slackness and 12.5 mm at the higher, both an order of magnitude worse than a well-chosen budget; the 4×200 configuration is the best accuracy-per-second in each; and the most accurate configuration lands at 1.3 mm regardless. That the ranking survives a doubling of slackness is evidence the guidance is about the solver rather than about one operating point. The table below is the $L/S = 2.55$ case, matching the photograph.

Every cable example shipped with Newton uses 10 substeps and 5 iterations. Applied to this problem that is far too coarse; but the repository’s previous default of 8×200 was tuned by varying iterations alone, leaving the substep axis untested. A sweep over both, scored against the analytic curve, shows accuracy to be essentially a function of the *product* — while cost is not, because each substep carries fixed overhead that an iteration does not. Fewer substeps with more iterations is therefore strictly cheaper at equal accuracy.

Separately, damping the stretch constraint proved actively harmful: it opposes the solver’s own length correction, and raising it to 10^{-2} cost roughly a factor of six in accuracy for no benefit. Damping the *bending* constraint, by contrast, changed nothing measurable over a $600\times$ range, because this cable’s EI is small enough that the bending mode carries almost no energy.

substeps	iterations	budget	RMSE [mm]	arc drift [%]	wall [s]
4	5	20	36.79	12.09	6.8
10	5	50	12.50	4.14	11.4
4	200	800	1.51	0.43	26.2
8	200	1600	1.23	0.29	203.7
10	200	2000	1.31	0.27	267.3
16	200	3200	1.64	0.23	433.9

Table 3: Selected configurations from the sweep.

The Warp rod and bending stiffness. One method needs a word of its own. The Warp rod is the only one that imposes a TRUE position-only pin at each support — the others weld the orientation too — and with that freedom its settled shape depends on the bending stiffness, the one input this project cannot measure. Parametrised by the elasto-gravitational length $\ell = (EI/w)^{1/3}$ and swept at two mesh resolutions (Figure 7), two things appear. First, accuracy degrades smoothly as the assumed stiffness rises: the stand-in material ($\ell \approx 150$ mm) is the worst, and the cable’s visible rounding scale (~ 2 cm, i.e. $\ell \approx 20$ mm) is far better. That ℓ is read from the rounding, deliberately not chosen to match the catenary — it lands well off it — which avoids the circular fit this method was written to avoid. Second, at the COARSE mesh the stiff case does not merely under-sag but buckles into an arch above the supports; refining the mesh removes the buckle and leaves the milder under-sag. So the dramatic buckle was partly a discretisation artefact, while the stiffness sensitivity is real and survives refinement. Either way the remedy is the same, and the Warp rod is a sound cable solver: the failures were an unphysically stiff assumed stiffness and too coarse a mesh, not the solver.

6 Provenance of the inputs

7 Validity

Frame calibration. The cable’s two ends are located geodesically — breadth-first search across the mask itself, twice — rather than as its leftmost and rightmost pixels. The distinction matters here: the tape clamps the cable’s *angle* as well as its position, so a slack cable leaves the clamp

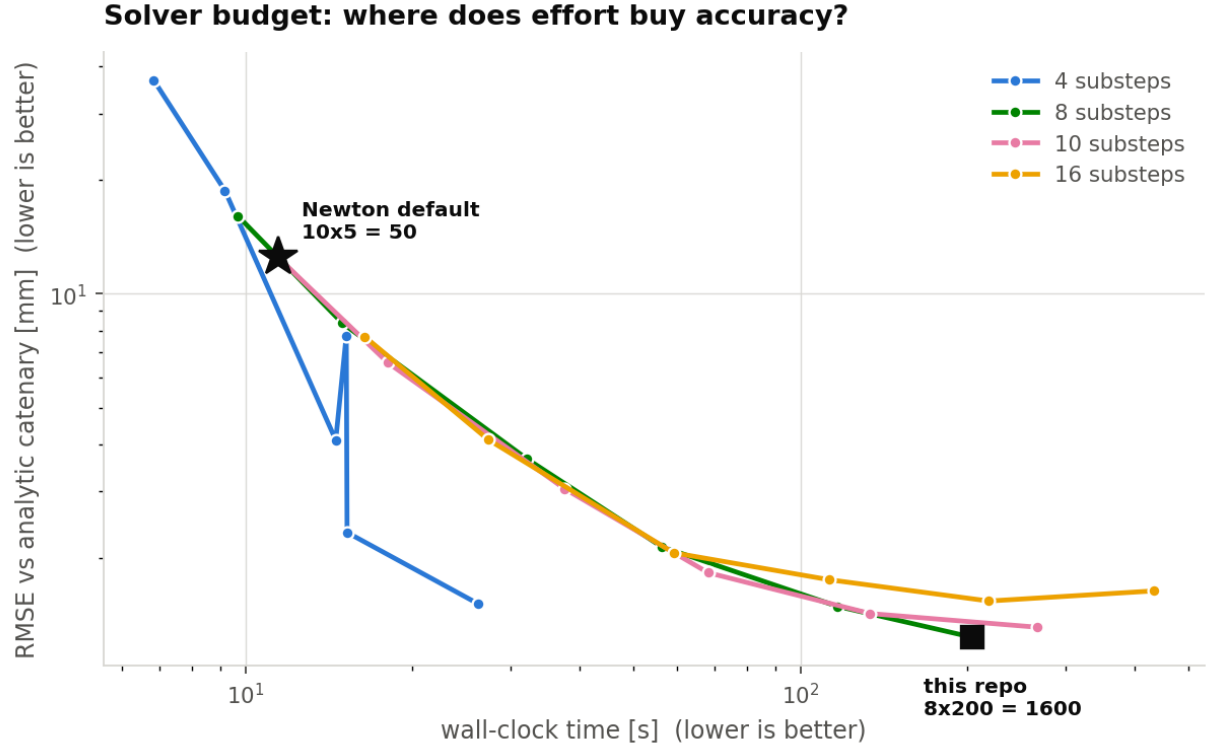


Figure 5: Accuracy against wall-clock cost. Both axes logarithmic; the knee is where further cost stops buying accuracy.

field	source	value
cable.length_m	measured	0.8932
supports.mid_fraction	measured	0.4688

Table 4: Which inputs were measured from the photograph and which were assumed. Recorded automatically for every run.

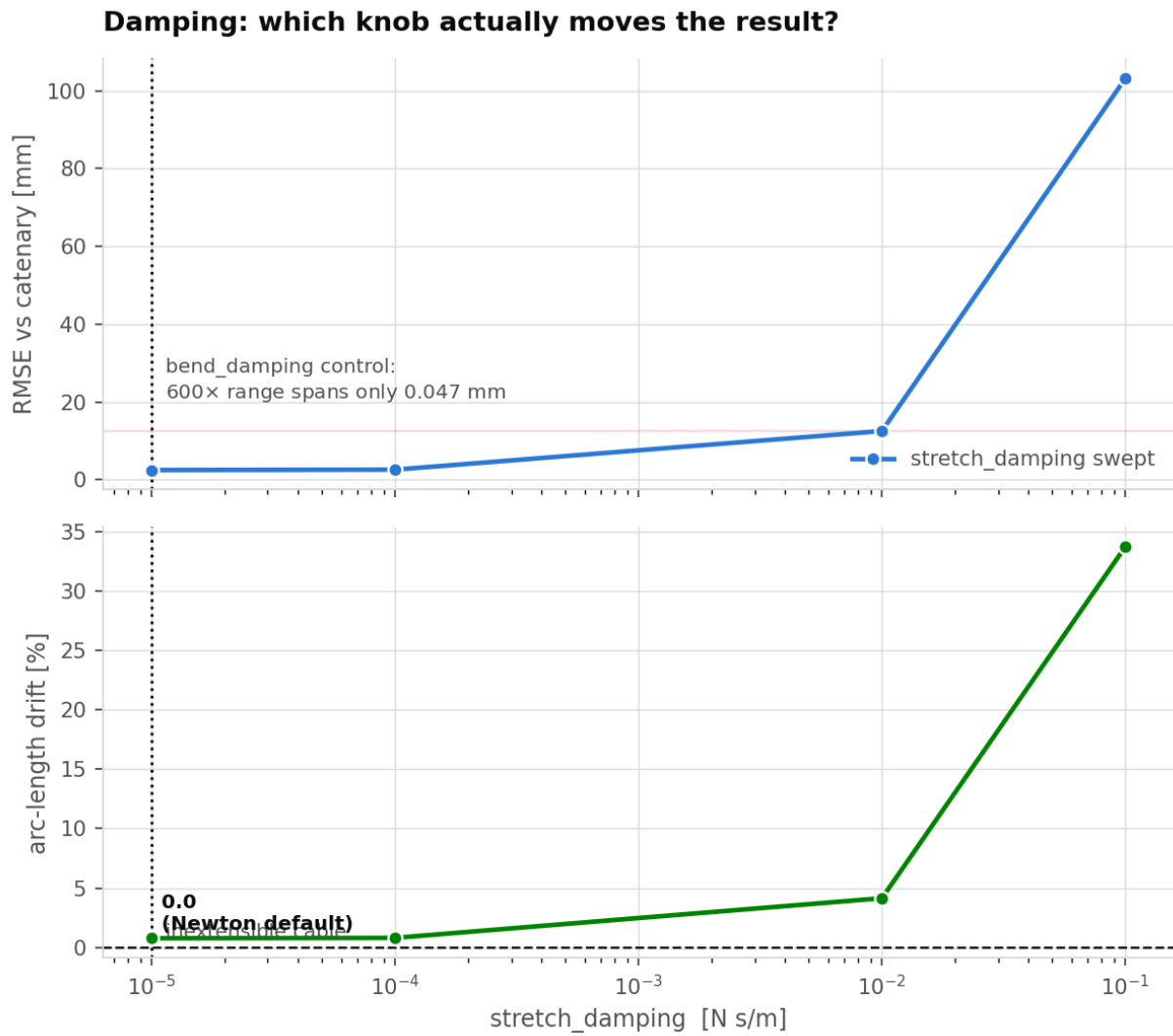


Figure 6: Damping. Stretch damping dominates; the bend-damping control spans a 600x range and does nothing.

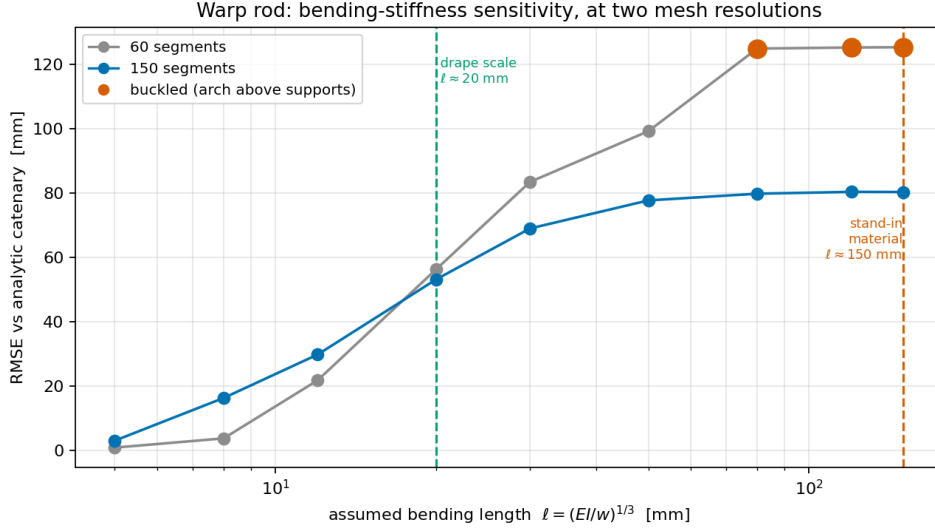


Figure 7: Warp rod accuracy against the assumed bending stiffness, parametrised by the elasto-gravitational length $\ell = (EI/w)^{1/3}$, at two mesh resolutions. Both curves rise with stiffness: the stand-in material ($\ell \approx 150$ mm) is worst, the cable’s ~ 2 cm rounding scale ($\ell \approx 20$ mm, used here) is far better and is read from the rounding, not fitted to the catenary. Only the coarse mesh buckles into an arch above the supports at high stiffness (orange); refining it leaves a milder under-sag — so the buckle was a discretisation artefact, the stiffness sensitivity is not.

along the tape’s direction and bends back, carrying it horizontally *past* its own anchor (86 px in this photograph). Taking the horizontal extreme therefore picks a point on the descending cable rather than the clamp.

An earlier version did exactly that at one end while the other came from where tracing halted — two ends defined by two different rules — and the resulting frame was tilted enough to place two physically level clamps 81 mm apart in height. With both ends found by one rule, and the clamp-to-clamp line adopted as the horizontal (the clamps are level in the apparatus, so this also removes camera roll, here -0.40°), that difference is now 0.0 mm by construction.

Independent check. The frame is fixed by the two OUTER clamps alone, so the middle clamp is free to disagree, and is therefore a genuine test. It is recovered at $x = 0.164$ m against 0.150 m measured by tape — 14 mm, or 4% of the span — and 3.9 mm above the outer clamps against 0 expected. Neither is forced by the calibration. The residual is consistent with hand-measuring the clamp positions and with lens distortion toward the frame edges, and it bounds the accuracy of any simulation-versus-photograph comparison below.

Independent reproducibility. The same physical cable was photographed a second time in a different rig — 2 clamps spanning 0.350 m rather than 3, a different camera orientation, and an independent scale calibration (193.109 vs 164.161 μm per pixel). Passed through the identical extraction, its free length comes out $L = 0.8929$ m against $L = 0.8932$ m here: a difference of 0.32 mm on a 893 mm cable, 0.036%. Two independent measurements of one object agreeing to a third of a millimetre is the strongest available evidence that the length the simulations are scored against is real and not an artefact of one photograph. The central inversion reproduces as well. The Newton cable method, which is the most faithful to the mathematics, reaches 1.51 mm against the analytic curve in the cross-check and 0.91 mm here, yet sits 71 mm and 75 mm from the two photographs respectively. The solver that nails the ideal curve stays far from the real cable in *both* geometries, so the gap is not an artefact of the middle clamp in the primary experiment — it appears already in the simplest single-span rig.

What the model-free check says. The traced arc length exceeds the length implied by fitting catenaries to the measured sags by roughly 58 mm. This is not numerical: it is the clamped end again. A catenary meets its support at whatever angle the length demands, while the tape forces a different angle and the cable bends to accommodate it, and that bend costs arc length the catenary does not spend. It is the largest single modelling discrepancy in this study and is a property of the apparatus, not of any solver.

Results that do not depend on the photograph. The following rest on the mathematics and the simulations alone, and are unaffected by any measurement made from the image:

- the verification of (1) against its closed form;
- every solver’s error against the analytic catenary;
- the hyperparameter sweep and its conclusions;
- the Warp rod’s stiffness sensitivity, and that its coarse-mesh buckle is a discretisation artefact the finer mesh removes (the *choice* of the floppier, drape-consistent stiffness uses the photograph, but these do not).

Modelling limits. The catenary is the $EI \rightarrow 0$ limit. The real cable is a charger cable with a permanent set, so its rest shape is not straight and some deviation from any ideal curve is physical rather than numerical. The tape also clamps the cable’s angle, which the classical catenary does not model. Both effects are concentrated near the clamps, and both are measurable, which makes them results to quantify rather than caveats to apologise for.

8 Conclusions

The solver that best reproduces the mathematics is not the one that best matches the photograph: Newton cable - point pin (kink) attains 0.91 mm against the analytic curve but 75.1 mm against the real cable, while PhysX capsule chain (D6) is the reverse (74.4 mm and 32.8 mm). This is the central result, and it is a statement about BOUNDARY CONDITIONS rather than about solver quality — see Section 8.

Why the ranking inverts. The photographed cable is held with tape, which fixes the direction in which the cable leaves the clamp. The classical catenary imposes no such condition: it meets its support at whatever angle the length happens to demand. These are different boundary-value problems on the same differential equation, and they have different solutions near the clamps.

Three independent observations support this reading rather than ”one solver is simply better”. First, the traced arc length exceeds the length implied by the measured sags by roughly 58 mm — extra cable spent bending to satisfy an imposed angle, which a catenary never spends. Second, the discrepancy is concentrated near the clamps and not distributed along the span. Third, the method that agrees most closely with the photograph is not the more accurate solver by any other measure; it is simply the one whose own error happens to lean the same way.

A solver that reproduces the analytic catenary to a millimetre is therefore behaving correctly. The residual against the photograph is a property of the apparatus, and closing it requires changing the MODEL — imposing the clamped end — not the solver or its settings.

What this argues for methodologically. With only simulation and photograph, the natural conclusion would have been that the most accurate solver was the least accurate, and effort would have gone into tuning a solver that was already right. The analytic leg is what distinguishes numerical error from modelling error, and it costs no experiment: it is available in closed form for exactly the idealisation the solvers claim to implement. That is the argument for carrying all three.

Sharpening the claim. It is not that the simulations use pinned ends and the apparatus uses clamped ones. Every method here *welds* its end bodies, so all of them impose a clamped end. The difference is the `ANGLE` at which it is clamped: the simulations start from the analytic catenary and therefore weld the ends at the catenary’s own exit angle, whereas the tape holds the real cable close to vertical. Same boundary condition type, different boundary value — and it is the value that the residual measures.

The test, carried out. The middle support admits the check directly. A strip of tape does not pin the cable at a point; it holds a short segment flat against the door, so the cable leaves the support horizontally and its bending stiffness rounds the top — whereas a point pin forces two catenary spans to meet with opposite slopes in an unavoidable upward kink (a catenary’s only horizontal tangent is its lowest point, and the middle support is a high one). Figure 3, inset, shows the photograph rounding over exactly as the tape model does and the point pin cornering. Replacing the pin with the flat-tape condition moves the Newton cable from 75 mm to 65 mm against the photograph while moving it from 0.9 mm to 21 mm against the analytic catenary. That is the signature the account predicts: the more faithful boundary condition is `FURTHER` from the idealised curve and `CLOSER` to the real cable. The residual is a model choice, and correcting the model closes part of it — without touching the solver or its settings.

The pinned-support check. The obvious check — rerun with a truly pinned rather than clamped middle support — cannot be done in Newton: its `add_rod` exposes capsule `BODIES` rather than nodes, and every way of holding a body also holds its orientation, so `--mid-support pin` returns a *bit-identical* result (same sag to 0.1 mm, same error to 0.01 mm) and a true position-only pin cannot be expressed through that API at all; the repository’s own `clamp_indices` documents this. The Warp rod *can* express it — a node of zero inverse mass constrains position alone — and once it is given a bending stiffness in the cable’s own floppy regime (Figure 7) it does exactly that: it pins the middle position, leaves both tangents free, and its bending then rounds the support just as the tape model and the photograph do (Figure 3, inset). That rounding, reached from a genuine position-only pin rather than an imposed flat clamp, is the independent confirmation Newton’s API cannot provide.

Extending the test to the outer clamps. The same idea applies at the two end supports, where the pinned-versus-clamped API limitation above does not obstruct it: impose the `MEASURED` end tangent rather than the catenary’s. The angle at which the cable leaves each outer clamp is directly readable from the photograph, and welding the end bodies at that angle would move the simulation further toward it. The residual would not close entirely — the cable’s permanent set, a charger cable’s memory of its coil, is a rest-shape effect no boundary condition can represent, and it is the most likely floor on any sim-to-photo agreement. Separating that floor from the remaining boundary-condition error is the natural next step; the middle-support result above shows the method works.